



FIELD PRACTICAL TRAINING REPORT

FPB 300

BOTSWANA UNIVERSITY OF AGRICULTURE AND NATURAL RESOURCES

BALAPOLOSI BOTLHE MORWAMANG

ATTACHMENT PLACEMENT: GREEN GLOBE (GLEN VALLEY)

IMPACT OF SALINITY IN CROPS AND SOIL IN GREEN GLOBE (GLEN VALLEY)



INTRODUCTION

The level of salt in a soil significantly affects crop growth and soil health. High salt levels increase osmotic pressure in the soil solution, making it harder for plants to absorb water leading to moisture stress (Maas & Hoffman, 1977). Excess sodium can also cause nutrient imbalance and delayed germination, prolonged salt build-up degrades soil causing soil compaction, crusting and poor drainage and reduced root penetration which further undermines soil productivity over time. (Richards, 1954).

The farm covers 9 hectares, with 5 hectares divided into four 1-ha blocks containing three shade nets and one open space plus a nursery tunnel, while the remaining 4 hectares are also divided into 1-ha blocks for open field production. This report evaluates the impact of soil salinity and sodium hazard on the crops and soil over multiple years.

METHODOLOGY

To determine how soil salinity affects crop growth and soil characteristics, a controlled experiment was set up. Soil tests measured salinity levels before planting and periodically during growth to monitor changes. Crops are planted under varying salinity conditions while keeping other factors constant such as watering, soil type and light.

A Hanna HI 9810 pH–EC–TDS meter was also used to conduct on-site soil salinity tests at Shade Net 1 and Open Space South which were the locations of focus as planting was to be done on those two locations. The practical skills are crucial for regular on-farm monitoring, enabling early detection of salinity trends and timely management interventions.

Year	Electric Conductivity	Sodium Absorption Ratio
2020	0.19	4.8
2021	0.22	5.3
2022	0.24	5.4
2023	0.26	5.5
2024	0.22	5.6

INTERPRETATION

Over the five year period, the electrical conductivity at different blocks of the farm shows a slight increase from 0.19 dS/m in 2020 to around 0.22 dS/m in 2024, indicating minor salt accumulation and the Sodium Adsorption Ratio increased moderately from 4.8 to 5.6, but it remains well below the critical threshold of 13 that would indicate salinity problems. Overall, these results suggest that the soil is currently safe for crops.

RESULTS

For comparison of soil salinity from the past years, soil was tested on Shade Net 1 and an Open Space North as the field was about to be planted on, hence testing the soil before transplanting, the soil samples were taken to Wellfield Consulting Services. Wellfield Consulting Service is well known to use their in-house laboratory to test soil, water and wastewater, offering chemical and trace element analysis

	Units	ELECTRICAL CONDUCTIVITY (dS/m)	Na Mg/l	Ca Mg/l	Mg Mg/l
SAMPLE A (SHADE NET 1)	Meq/l	0.2234	230	72	37
OPEN SPACE SOUTH	Meq/l	0.2218	230	84	24

Sodium Absorption Ratio (SAR)

It is a measurement used to assess the sodium hazard of irrigation water or soil extract. It tells you how much sodium (Na^+) is present relative to calcium (Ca^{2+}) and magnesium (Mg^{2+}).

First calculate the Meq/l

$$\text{Mg/l} = \frac{\text{mg/l}}{\text{Equivalent weight}} \times \text{Valency}$$

$$\text{Na} = 23/1 = 23$$

$$\text{Mg} = 24/2 = 12$$

$$\text{Ca} = 40/2 = 20$$

	Units	ELECTRICAL CONDUCTIVITY (dS/m)	Na Meq/l	Ca Meq/l	Mg Meq/l
SAMPLE A (SHADE NET 1)	Meq/l	0.2234	10	3.6	3.1
SAMPLE B (OPEN SPACE SOUTH)	Meq/l	0.2218	10	4.2	2

$$SAR = \frac{[Na^+]}{\sqrt{([Ca^{2+}] + [Mg^{2+}]) / 2}}$$

$$\sqrt{([Ca^{2+}] + [Mg^{2+}]) / 2}$$

SAMPLE A

$$SAR = \frac{10}{\sqrt{(3.6 + 3.1)/2}}$$

$$\sqrt{(3.6 + 3.1)/2}$$

$$= 5.5$$

SAMPLE B

$$SAR = \frac{10}{\sqrt{(4.2 + 2)/2}}$$

$$\sqrt{(4.2 + 2)/2}$$

$$= 5.7$$

Electric Conductivity indicates the salinity in the soil or irrigation water which prevents crop growth.

Sample A shows that there is low salinity indicating low sodium hazard.

Sample B shows that there is low salinity indicating low sodium hazard.



For hands-on experience, a Hanna HI 9810 pH–EC–TDS meter was used on-site to test soil salinity levels in Shade Net 1 and Open Space South. This portable device allowed for immediate field measurements of electrical conductivity and total dissolved solids which would be used to indicate the sodium adsorption ratio. The field readings closely matched the laboratory results from Wellfield Consulting Services confirming the reliability of both methods.

The analysis revealed that Shade Net 1 recorded sodium adsorption ratio of 5.5, while the Open Space South recorded a slightly higher sodium adsorption ratio of 5.7. Both values fall well below the critical threshold of 13, indicating low sodium hazard and minimal risk of soil structure degradation. The difference in exposure to rainfall and evaporation between the two areas indicates that open spaces tend to accumulate more salts due to higher evaporation rates compared to shaded environments.

DISCUSSION



The tested soil samples show that both sampled areas at Shade Net 1 and an Open Space South have low salinity levels based on electrical conductivity values below 0.5 dS/m and Sodium Absorption Ratio values below 6 and this indicates a favorable soil environment for most crops, with minimal risk of salinity.

The electrical conductivity (EC) values for both samples (0.22 dS/m) fall well below the threshold of 2 dS/m, which is the general limit above which sensitive crops may begin to experience reduced yields. This suggests that soluble salt concentrations are not high enough to interfere with water uptake or cause osmotic stress, and the soil can support healthy crop growth under normal irrigation practices.

In terms of Sodium Adsorption Ratio (SAR), the values of 5.5 (Shade Net 1) and 5.7 (Open Space South) further indicate that sodium is not high enough to cause a salinity hazard. Salinity soils, typically characterized by sodium adsorption ratio greater than 13 which can lead to the breakdown of soil structure, reduced infiltration and poor aeration. The absence of such conditions in the tested plots means that soil structure and permeability are likely intact, which supports good root development and water movement (Ayers & Westcot, 1985).

CONCLUSION

The soil salinity assessment revealed that both sampled areas, indicate low salinity levels as shown by electrical conductivity (EC) values consistently below 0.25 dS/m and Sodium Adsorption Ratio (SAR) values under 6. These results indicate that the soils are safe for crop production with minimal risk of osmotic stress or sodium induced structural problems.

Low electrical conductivity (EC) values confirm that soluble salt concentrations in the soil are not high enough to interfere with water uptake by crops, while low sodium adsorption ratio (SAR) values suggest that soil structure remains stable and infiltration rates are not interfered. Hence, this indicates healthy root development, good nutrient availability and sustained crop growth throughout the production cycle.

RECOMMENDATIONS

- Regular monitoring by testing soil and the soil adsorption ratio annually to detect early salt buildup.
- Leaching irrigation by applying occasional deep irrigation during rainy seasons to flush excess salts.
- Balanced fertilization in order to avoid over-application of sodium based fertilizers.
- Mulching by the use of organic mulches in open spaces to reduce surface evaporation and salt concentration.
- Use of quality irrigation water by regularly testing irrigation water for salinity and to ensure it does not contribute to long-term salt buildup.
- Training and awareness by providing training to farm workers on recognizing early signs of salinity stress in crops and appropriate management practices.

REFERENCE

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